FIITJEE

ALL INDIA TEST SERIES

FULL TEST – X

JEE (Main)-2023

TEST DATE: 31-03-2023

Time Allotted: 3 Hours **Maximum Marks: 300**

General Instructions:

- The test consists of total 90 questions.
- Each subject (PCM) has 30 questions.
- This question paper contains **Three Parts.**
- **Part-A** is Physics, **Part-B** is Chemistry and **Part-C** is Mathematics.
- Each part has only two sections: **Section-A and Section-B**.
- **Section – A** : Attempt all questions.
- **Section – B :** Do any five questions out of 10 questions.

  Section-A (01 – 20, 31 – 50, 61 – 80) *contains 60 multiple choice questions which have* ***only one correct answer****. Each question carries* ***+4 marks*** *for correct answer and* ***–1 mark*** *for wrong answer.*

  Section-B (21 – 30, 51 – 60, 81 – 90) *contains 30 Numerical based questions. The answer to each question is rounded off to the nearest integer value. Each question carries* ***+4 marks*** *for correct answer and* ***–1 mark*** *for wrong answer.*

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Physics

PART – A

SECTION – A

(Single Choice Answer Type)

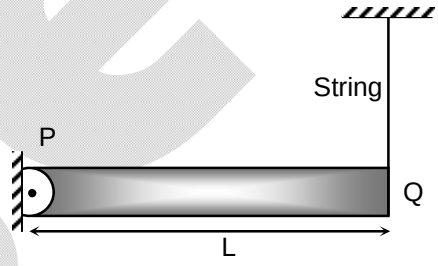
This section contains **20 multiple choice questions**. Each question has four choices (A), (B), (C) and (D) out of which **ONLY ONE is correct.**

1. A satellite of mass m orbits the earth in an elliptical orbit having aphelion distance r_a and perihelion distance r_p. The period of the orbit is T. The semi-major and semi-minor axes of the ellipse are $\frac{r_a + r_p}{2}$ and $\sqrt{r_p r_a}$ respectively. The angular momentum of the satellite is

(A) $\frac{m\pi(r_a + r_p)\sqrt{r_a r_p}}{T}$ (B) $\frac{2m\pi(r_a + r_p)\sqrt{r_a r_p}}{T}$

(C) $\frac{m\pi(r_a + r_p)\sqrt{r_a r_p}}{2T}$ (D) $\frac{m\pi(r_a + r_p)\sqrt{r_a r_p}}{4T}$

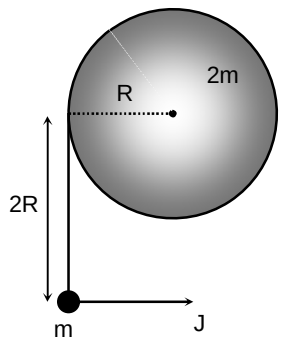
2. A uniform rod PQ of length L is hinged at one end P. The rod is kept in the horizontal position by a massless string tied to point Q as shown in the figure. If the string is cut, the initial angular acceleration of the rod will be



(A) $\frac{g}{L}$ (B) $\frac{2g}{L}$

(C) $\frac{2g}{3L}$ (D) $\frac{3g}{2L}$

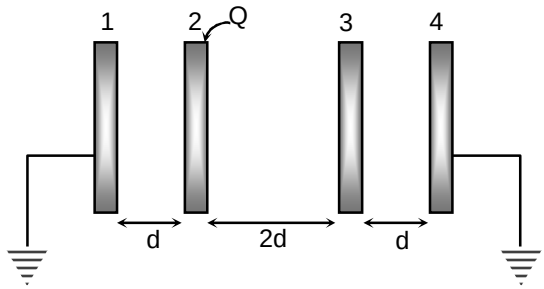
3. A uniform circular disc of mass 2m and radius R is placed freely on a horizontal smooth surface. A particle of mass m is connected to the circumference of the disc with a massless string. Now an impulse J is applied on the particle in the direction as shown in the figure. If J = 10 Ns, $m = \sqrt{10}$ kg and R = 25 cm, then acceleration of centre of mass of the disc just after application of impulse is



(A) 1 m/s^2 (B) 2 m/s^2

(C) 3 m/s^2 (D) 4 m/s^2

4. Four large identical metallic plates are placed as shown in the figure. Plate-2 is given a charge Q. All other plates are neutral. Now plate-1 and plate 4 are earthed. Area of each plate is A. Then charge on right surface of plate 3 is

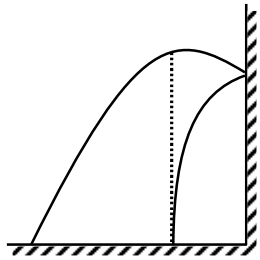


(A) $\frac{Q}{3}$ (B) $\frac{Q}{2}$

(C) $\frac{Q}{4}$ (D) $-\frac{Q}{3}$

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5. A stone is projected from a horizontal plane. It attains maximum height H and strikes a stationary smooth wall and falls on the ground vertically below the maximum height. Assume the collision to be elastic, the height of the point on the wall where stone will strike is



(A) $\frac{H}{2}$ (B) $\frac{H}{4}$

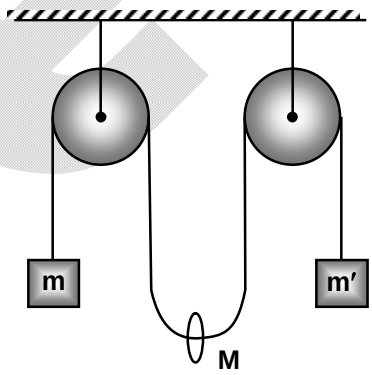
(C) $\frac{3H}{4}$ (D) $\frac{H}{3}$

6. A smooth sphere is moving on a horizontal surface with velocity vector $2\hat{i}+2\hat{j}$ immediately before it hits a vertical wall. The wall is parallel to $\hat{j}$ vector and the coefficient of restitution between the sphere and the wall is $\frac{1}{2}$. The velocity vector of the sphere after it hits the wall is

(A) $\hat{i}-\hat{j}$ (B) $-\hat{i}+2\hat{j}$

(C) $-\hat{i}-\hat{j}$ (D) $2\hat{i}-\hat{j}$

7. A smooth ring of mass M is threaded on a string as shown in the figure. Various portions of string are vertical. If the ring alone is to remain at rest, then



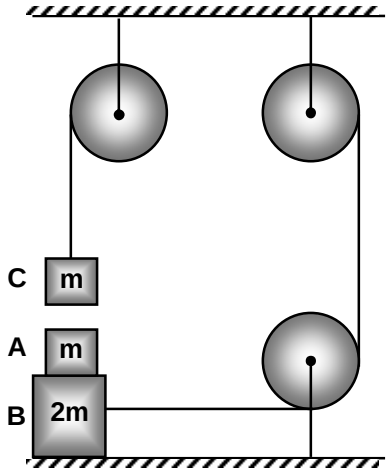
(A) $\frac{4}{M}=\frac{1}{m}+\frac{1}{m'}$

(B) $\frac{2}{M}=\frac{1}{m}+\frac{1}{m'}$

(C) $\frac{1}{M}=\frac{1}{m}+\frac{1}{m'}$

(D) $\frac{3}{M}=\frac{1}{m}+\frac{1}{m'}$

8. In the arrangement shown in the figure. There is friction force between the block of masses m and 2m. Block of mass 2m is kept on a smooth horizontal plane. The mass of suspended block is m. Block A is stationary with respect to block of mass 2m. The minimum value of coefficient of friction between m and 2m is s

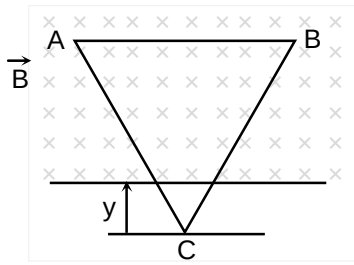


(A) $\frac{1}{2}$ (B) $\frac{1}{\sqrt{2}}$

(C) $\frac{1}{4}$ (D) $\frac{1}{3}$

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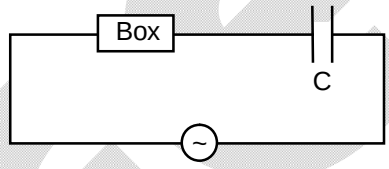
9. A current carrying loop is in the shape of an equilateral triangle of side length a. Its mass is M and it is in vertical plane. There exist a uniform horizontal magnetic field B is the region shown. The loop is in equilibrium for $y = \frac{\sqrt{3}a}{4}$. Then current in the loop is



(A) $\frac{Mg}{aB}$ (B) $\frac{2Mg}{aB}$

(C) $\frac{Mg}{2aB}$ (D) $\frac{\sqrt{3}Mg}{2aB}$

10. A box has a large electric current inside it. When it was connected to an ac generator it was found that it was putting a lot of load on the generator and the power factor of the box was $\frac{1}{\sqrt{2}}$. A capacitor of capacitance C was connected in series with the box and the power factor of the circuit became equal to the ideal value. Then the impedance of the box (if generator has angular frequency of ω) is



(A) $\frac{1}{\omega C}$ (B) $\frac{2}{\omega C}$

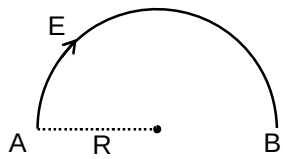
(C) $\frac{\sqrt{2}}{\omega C}$ (D) $\frac{1}{2\omega C}$

11. An ideal gas is made to under go a process $T = T_0 e^{\alpha V}$, where T_0 and α are constants. The molar specific heat capacity of the gas in the process if its molar specific heat capacity at constant volume is C_V and molar specific heat capacity at constant pressure is C_P, is

(A) $C_V + \frac{\alpha R}{V}$ (B) $C_P + \frac{\alpha R}{V}$

(C) $C_V + \frac{R}{\alpha V}$ (D) $C_P + \frac{R}{\alpha V}$

12. In a region of space an electric field line is in the shape of semicircle of radius R. Magnitude of the field at all points is E. A particle of mass m having charge q is constrained to move along this field line. The particle is released from rest at A. The acceleration of the particle when it is at midpoint of path from A to B.

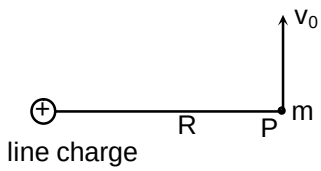


(A) $\frac{qE}{m}$ (B) $\frac{qE}{m}\pi$

(C) $\frac{qE}{m}\sqrt{1+\pi^2}$ (D) 0

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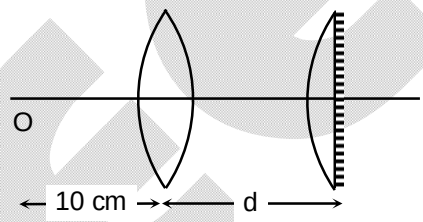
13. An infinite line charge is perpendicular to the plane of paper having linear charge density λ. A particle having charge Q and mass m is projected in the field of the line charge from point P. The point P is at a distance R from the charge and velocity given to the particle is perpendicular to the radial line at P. The speed of particle when it is at distance nR, where n>1, from the line charge is



(A) $\sqrt{V_0^2+\frac{\lambda Q}{\pi\varepsilon_0 m}\ell n 2n}$ (B) $\sqrt{V_0^2+\frac{2\lambda Q}{\pi\varepsilon_0 m}\ell n\, n}$

(C) $\sqrt{V_0^2+\frac{\lambda Q}{\pi\varepsilon_0 m}\ell n\, n}$ (D) none of these

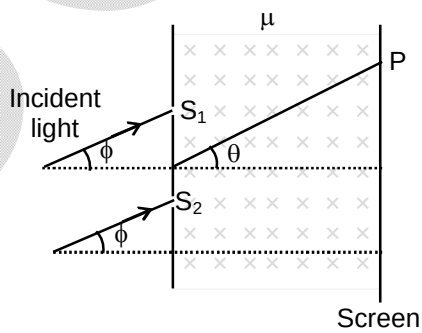
14. A convex lens of focal length 20 cm and another plano-convex lens of focal length 40 cm are placed co-axially. The plane surface of the plano-convex lens is silvered. An object O is kept on the principal axis at a distance of 10 cm from the convex lens. Find the distance d between the two lenses so that the final image is formed on the object itself.



(A) 10 cm (B) 30 cm

(C) 20 cm (D) 40 cm

15. Light is incident at an angle ϕ with the normal to a vertical plane containing 2 narrow slits (S_1 and S_2) at separation d. The medium to the left of slit plane is air and wavelength of the incident light is λ. The medium to the right of the slit plane has refractive index μ. The value of angular position (θ) of a point P where we will observe constructive interference is



(A) $\sin^{-1}\left(\frac{n\lambda-d\sin\phi}{\mu d}\right), n\in I$ (B) $\cos^{-1}\left(\frac{n\lambda-d\sin\phi}{\mu d}\right) n\in I$

(C) $\sin^{-1}\left(\frac{n\lambda/2-d\sin\phi}{\mu d}\right) n\in I$ (D) $\cos^{-1}\left(\frac{\frac{n\lambda}{2}-d\sin\phi}{\mu d}\right) n\in I$

16. Rate of evaporation (volume evaporated per unit time) of water kept in an earthen pot is proportional to the volume of water present in the pot. It is observed that it takes 48 hrs for 75% of the water kept in the pot to evaporate. The empting pot was placed below a tap from which water leaks in similar drops each of volume V_0. The drops fall at a uniform rate of n drops per hour. The volume of water in the pot 24 hours after it was kept below the tap is

(A) $\frac{6}{\ln 2}nv_0$ (B) $\frac{8}{\ln 2}nv_0$

(C) $\frac{10}{\ln 2}nv_0$ (D) $\frac{12}{\ln 2}nv_0$

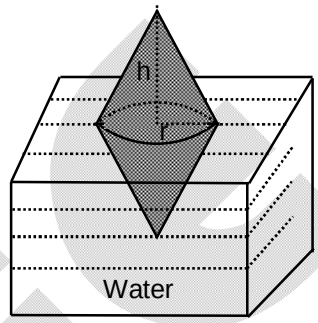
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17. If intensity of central maxima in Fraunhoffer's diffraction pattern is I_0, intensity of second secondary maxima is (approximately)

(A) I_0 (B) $\frac{I_0}{4}$

(C) $\frac{I_0}{22}$ (D) $\frac{I_0}{61}$

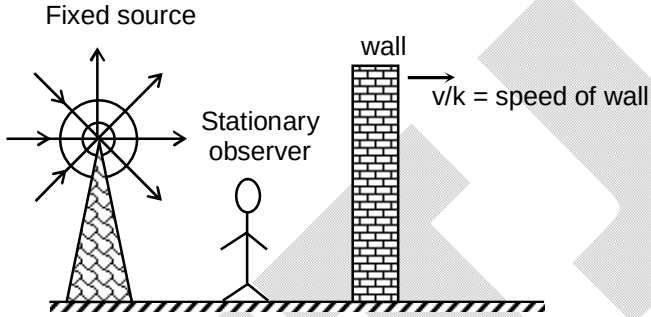
18. The figure here shows a composite solid body (made up by joining two identical cones of height h) floating half submerged in water. Angular frequency of oscillation of the body, on slight vertical displacement from shown position is



(A) $\sqrt{\frac{g}{h}}$ (B) $\sqrt{\frac{2g}{h}}$

(C) $\sqrt{\frac{3g}{h}}$ (D) $2\sqrt{\frac{g}{h}}$

19. If frequency of sound emitted by the source is f, then beat frequency observed by the stationary observer is (v = speed of sound)

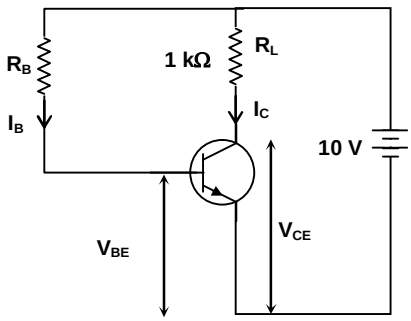


(A) $\frac{f}{k}$ (B) $\frac{f}{k+1}$

(C) $\frac{2f}{k+1}$ (D) $\frac{2kf}{k+1}$

20. In the circuit shown in figure the current gain β = 100 for an n-p-n transistor. The bias resistance R_B, so that V_{CE} = 5 V is (V_{BE} << 10 V)

(A) $2 \times 10^3\ \Omega$
(B) $10^5\ \Omega$
(C) $2 \times 10^5\ \Omega$
(D) $4 \times 10^5\ \Omega$



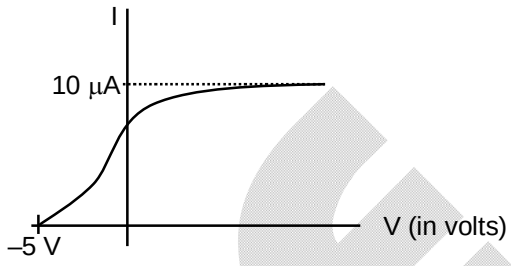
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SECTION – B

(Numerical Answer Type)

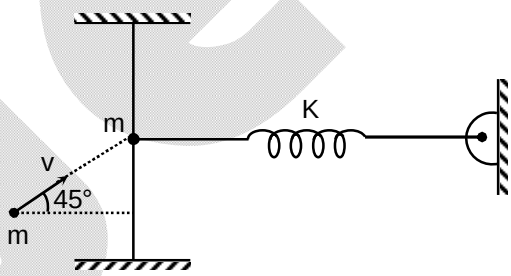
This section contains **10** Numerical based questions. The answer to each question is rounded off to the nearest integer value.

21. In a photoelectric experiment a metal plate of work function ϕ = 2 eV is irradiated with beam of monochromatic light. The photoelectric current (I) vs the applied potential difference (V) graph is as shown in the figure. It is known that the efficiency of photo emission is 0.5×10^{-3} %. then the power of light incident an metal plate (in watts) is

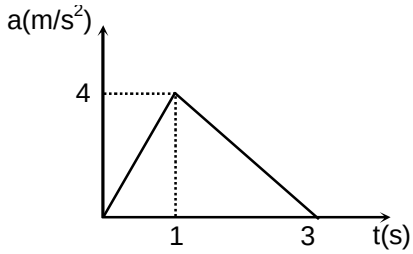


22. The mean life of ^{55}Co if its activity is known to decrease 4% per hour (in hours) is (Given $\ell n\left(\frac{25}{24}\right) = 0.04$)

23. The figure here shows a bead of mass m = 1 kg attached to a relaxed spring of length 80 cm and spring constant k = 100 N/m. The bead is constrained to slide on a smooth long vertical rod in a gravity free space. A particle of mass 1 kg moving with v = 4 m/s at 45° with horizontal strikes the bead and sticks to it. The maximum displacement of bead (in cm) w.r.t. initial position is

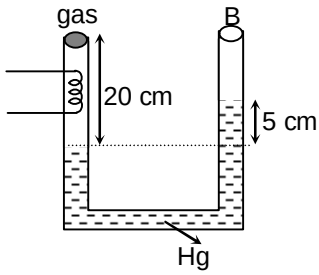


24. The acceleration time graph of a particle moving in a straight line is as shown in the figure. The velocity of the particle at time t = 0 is 2 m/s. The velocity after 2 second (in m/s)



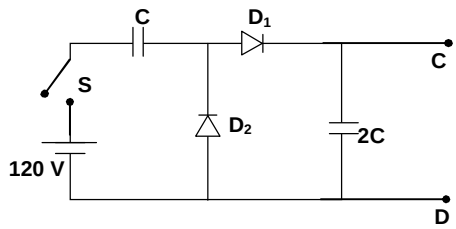
25. A cylinder of oxygen, used as medical aid has a volume of 16 liters. It is filled to a pressure of 1.49×10^7 N/m^2 above the atmospheric pressure at room temperature of 300 K. When in use, the flow rate at atmospheric pressure is 2.4 L/min. The cylinder lasts for 10^x minutes. The value of x is (Atmospheric pressure is 10^5 N/m^2)

26. One end of an insulating U-tube is sealed using insulating material. A monoatomic gas at T = 300 K occupies 20 cm length of tube as shown in the figure. The level of mercury on two sides of the tube differs by 5 cm. The other end of the tube is open to atmosphere. Area of cross-section of the tube is uniform and is equal to 0.01 m^2. The gas in the tube is heated by an electric heater so as to raise its temperature to 562.5 K. Assume no heat is conducted to mercury by gas. The final length of gas column (in cm) is (assume atmospheric pressure is equivalent to 75 cm of Hg column in barometer tube)

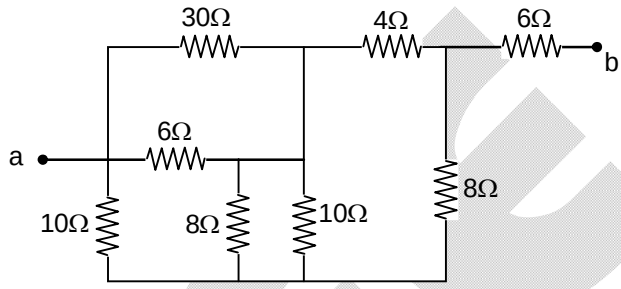


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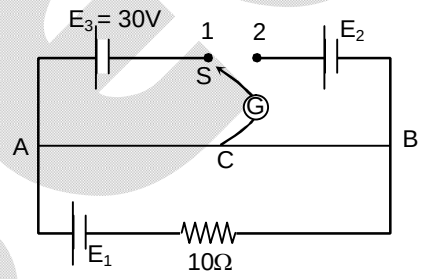
27. A diode is a device that conducts in one direction only. The potential difference (in volts) between terminals C and D after switch (S) is closed in the circuit shown in the figure is



28. Equivalent resistance between points a and b in the network in Ω is



29. In the circuit shown, AB is a uniform wire of length L = 5m. It has a resistance of 2 Ω/m. When AC = 2m, it was found that the galvanometer shows zero reading when S is placed in either of the two position 1 or 2. Then the emf E_1 (in volts) is



30. A concave mirror forms a real image on a screen of thrice the linear dimension of a real object placed perpendicular to its principal axis. The mirror is moved by 10 cm along its principal axis and once again a sharp image of the object is formed on the screen. This time the image is twice as large as the object. The focal length of the mirror (in cm) is

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Chemistry

PART – B

SECTION – A

(Single Choice Answer Type)

This section contains **20 multiple choice questions**. Each question has four choices (A), (B), (C) and (D) out of which **ONLY ONE is correct.**

31. Which of the following statement is not true for the reaction given below?

$\left[Cu(H_2O)\right]^{2+} + 4NH_3 \rightleftharpoons \left[Cu(NH_3)_4\right]^{2+} + 4H_2O$

(A) It is ligand substitution reaction
(B) NH_3 is relatively strong field ligand while H_2O is weak field ligand
(C) During the reaction, there is a change in colour form light blue to dark blue
(D) $\left[Cu(NH_3)_4\right]^{2+}$ has tetrahedral structure and is paramagnetic

32. When ZnS and PbS minerals are present together, then NaCN is added to separate them in the froth floatation process as a depressant, because:

(A) $Pb(CN)_2$ is precipitated while no effect on ZnS
(B) ZnS forms soluble complex $Na_2\left[Zn(CN)_4\right]$
(C) PbS forms soluble complex $Na_2\left[Pb(CN)_4\right]$
(D) They cannot be separated by adding NaCN

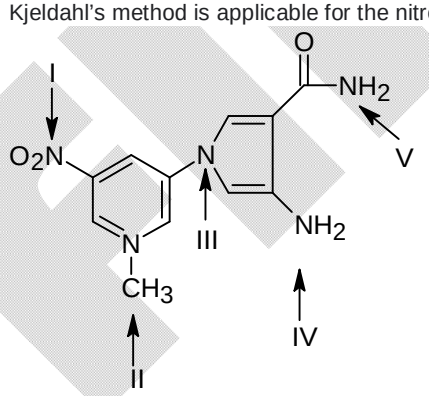
33. Which of the following metal on burning in moist air does not give smell of ammonia?

(A) Mg (B) Ca
(C) K (D) Li

34. Which is not true about borax?

(A) It is a useful primary standard for titrating against acids
(B) One mole of borax contains 4B – O – B bonds
(C) Aqueous solution of borax can be used as buffer
(D) It is made up of two triangular BO_3 units and two tetrahedral BO_4 units

35. Kjeldahl's method is applicable for the nitrogens as indicated



(A) All except II and III (B) All except III
(C) IV and V only (D) I and IV only

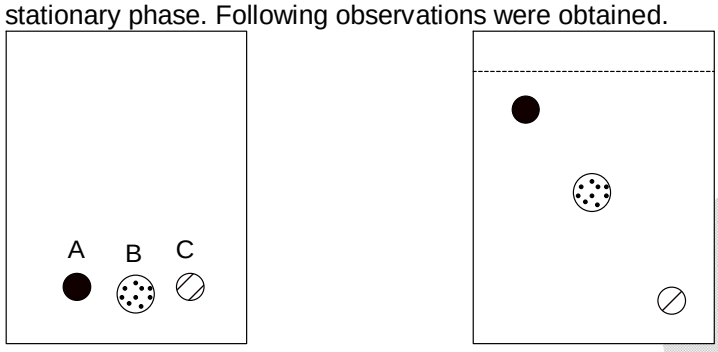
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36. A sulphate of a metal (A) on heating evolves two gases (B) and (C) and an oxide (D). Gas (B) turns $K_2Cr_2O_7$ paper green while gas (C) forms a trimer in which there is no S – S bond. Compound (D) with conc. HCl forms a Lewis acid (E) which exists in a dimer. Compounds (A), (B), (C), (D) and (E) are respectively:

(A) $FeSO_4, SO_2, SO_3, Fe_2O_3, FeCl_3$ (B) $Al_2(SO_4)_3, SO_2, SO_3, Al_2O_3, FeCl_3$

(C) FeS, SO_2, SO_3, $FeSO_4$, $FeCl_3$ (D) FeS, SO_2, SO_3, $Fe_2(PO_4)_3$, $FeCl_2$

37. Mixture of A, B and C was analysed TLC technique using a non-polar solvent with a polar stationary phase. Following observations were obtained.



Before After

Statements about the above experiments are

I. Polarity of A, B and C is in order A > B > C

II. R_f value of A, B and C is in order A > B > C

III. R_f value of A, B and C is index C > B > A

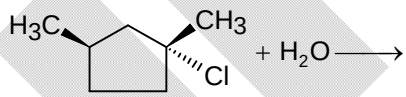
IV. C is most polar and A is most non-polar

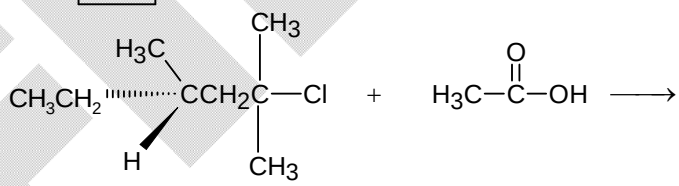
(A) I, II (B) I, III

(C) I, IV (D) II, IV

38. A pair of diastereomers is formed in the following reaction:

(A) 

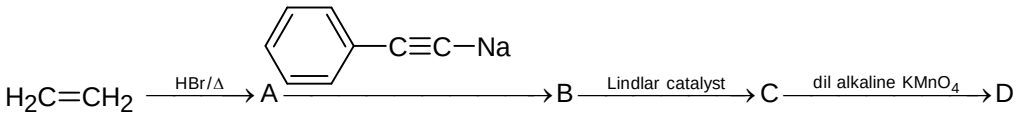
(B) 

(C) 

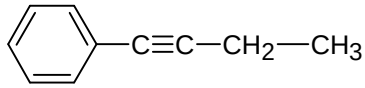
(D) All of these

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39. Based on the following sequence of reactions, select incorrect statements



(A) A is primary alkyl bromide

(B) B is 

(C) C is a cis-isomer

(D) D is obtained by anti addition

40. Both $N(SiH_3)_3$ and $NH(SiH_3)_2$ compounds have trigonal planar skeleton. Incorrect statements about both compounds is:

(A) Si-N-Si bond angle in $NH(SiH_3)_2$ > SiNSi bond angle in $N(SiH_3)_3$

(B) N – Si bond length in $NH(SiH_3)_2$ > N – Si bond length in $N(SiH_3)_3$

(C) N – Si bond length in $NH(SiH_3)_2$ < N – Si bond length in $N(SiH_3)_3$

(D) Back bonding strength in $NH(SiH_3)_2$ > Back bonding strength in $N(SiH_3)_3$

41. The correct increasing order of extent of hydrolysis is:

(A) $CCl_4 < MgCl_2 < AlCl_3 < SiCl_4 < PCl_5$

(B) $CCl_4 < AlCl_3 < MgCl_2 < PCl_5 < SiCl_4$

(C) $CCl_4 < SiCl_4 < PCl_5 < AlCl_3 < MgCl_2$

(D) $CCl_4 < PCl_5 < SiCl_4 < AlCl_3 < MgCl_2$

42. $CuSO_4\,(aq.) \xrightarrow{\text{Excess of KCN}} N + O$

Then final products N and O are respectively

(A) $\left[Cu(CN)_4\right]^{3-}, (CN)_2$

(B) $CuCN, (CN)_2$

(C) $\left[Cu(CN)_4\right]^{2-}, (CN)_2$

(D) $Cu(CN)_2, K_2S$

43. Which of the following reaction(s) give oxo-acid?

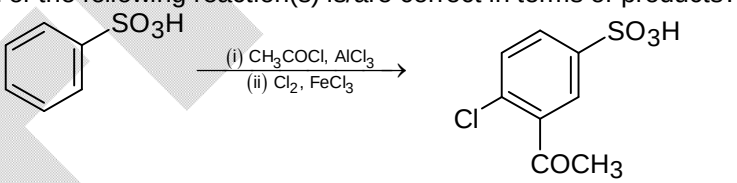
(A) Two moles of nitric acid $\xrightarrow{-H_2O}$

(B) One mole of sulphurus acid $\xrightarrow{-H_2O}$

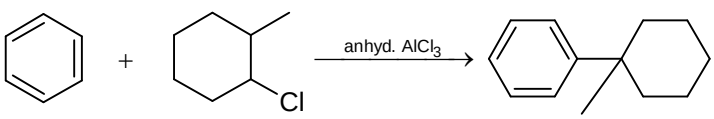
(C) Two moles of chloric acid $\xrightarrow{-H_2O}$

(D) Two moles of sulphuric acid $\xrightarrow{-H_2O}$

44. Which of the following reaction(s) is/are correct in terms of products?

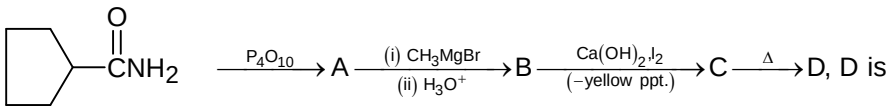
(A) 

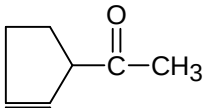
(B) 

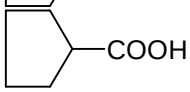
(C) 

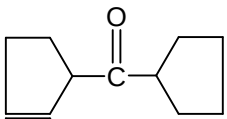
(D) All of these

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45. 

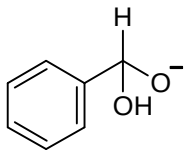
(A) 

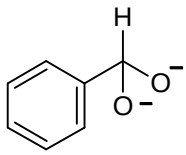
(B) 

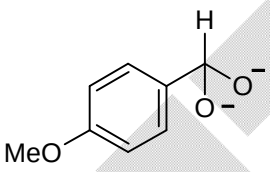
(C) 

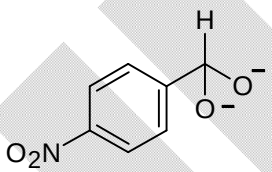
(D) 

46. In Cannizzaro reaction, the intermediate that will be the best hydride donor is

(A) 

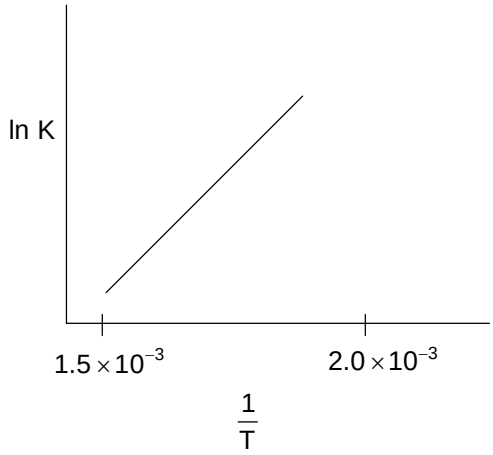
(B) 

(C) 

(D) 

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47. The graph relates $\ln K_{eq.}$ vs. $\frac{1}{T}$ for a reaction, the reaction must be:



(A) exothermic
(B) endothermic
(C) ΔH is negligible
(D) spontaneous

48. Which statement wrong about $R-\underset{\underset{NH_2}{|}}{CH}-COOH$, if the following equilibrium exist

$$R-\underset{\underset{NH_3^+}{|}}{CH}-COOH \overset{K_{a_1}}{\rightleftharpoons} R-\underset{\underset{NH_3^+}{|}}{CH}-COO^- \overset{K_{a_2}}{\rightleftharpoons} R-\underset{\underset{NH_2}{|}}{CH}-COO^- + H^+$$

Conjugate acid of Zwitter ion — Zwitter ion — Conjugate base of Zwitter ion

(A) $\left[H^+\right] = \sqrt{K_{a_1}.K_{a_2}}$

(B) $pH = \frac{pK_{a_1} + pK_{a_2}}{2}$

(C) The pH of aqueous solution depends upon concentration of $R-\underset{\underset{NH_2}{|}}{CH}-COOH$

(D) The concentration of $[H^+]$ for Zwitter ion can be calculated for any amphoteric salt such as HCO_3^-

49. Which of the following are not the derivatives of barituric acid?
(A) Amytal
(B) Luminal
(C) Seconal
(D) Mapromate

50. Which of the following size of particulate will cause diseases related to lungs?
(A) 10^{-2} meter
(B) 10^{-6} meter
(C) 10^{-4} meter
(D) 10^{-1} meter

SECTION – B

(Numerical Answer Type)

This section contains **10** Numerical based questions. The answer to each question is rounded off to the nearest integer value.

51. Consider the cyclic trimer structure of SO_3. Calculate the value of p + q. Here p and q are total number of $d\pi - p\pi$ bond and total number of sp hybidised atom respectively.

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52. There are some species given below

(i) O_2^+

(ii) CO

(iii) B_2

(iv) O_2^-

(v) NO^+

(vi) He_2^+

(vii) C_2^{2+}

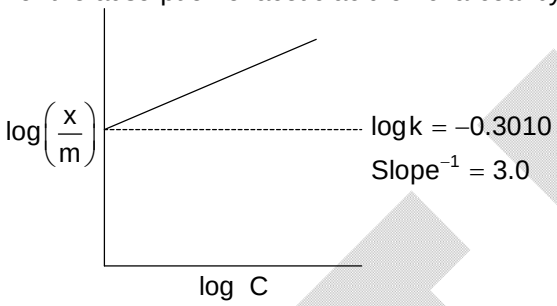
(viii) CN^-

(ix) N_2

Total number of species which have their fractional bond order.

53. 1 mole of phenol and 1 mole of benzoic acid on reaction with $NaHCO_3$ at 300 K and 1 bar forms …. X dm^3 CO_2. Thus, X is……

54. For the adsorption of acetic acid on charcoal by Freundlich adsorption isotherm at 298 K



55. K_a for HCN is 5×10^{-10} at 25°C. For maintaining a constant pH of 9, the volume of 5 M KCN solution required to be added to 10 mL of 2 M HCN solution is…

56. Consider following reactions I and II with their kinetics parameter at 600 K

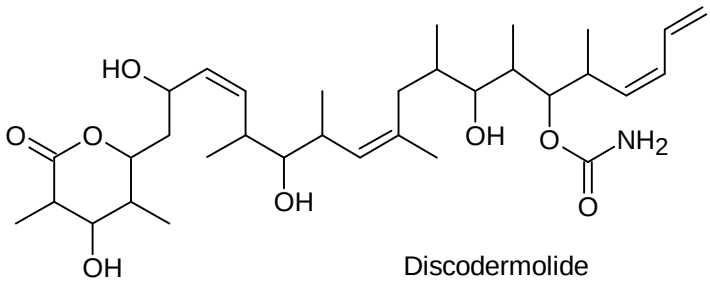
	Reaction	Rate constant k	Activation energy
I.	$HCOOH \rightarrow H_2 + CO_2$	11.5×10^{-4} s^{-1}	30 kJ mol^{-1}
II.	$HCOOH \rightarrow H_2O + CO$	3.0×10^{-4} s^{-1}	15 kJ mol^{-1}

Temperature at which equimolar quantities of H_2O, CO, CO_2 and H_2 are formed is ……K.

57. How many following metals evolve NO (Nitric oxide) gas with dil. HNO_3 (20%)?
Hg, Cu, Pb, Zn, Fe, Al, Ag, Au, Mn

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58. Considering all stereogenic centres, what is maximum number of stereoisomers possible for discodermolide?



Discodermolide

Express results in the form of 2^x. Thus, x = ……

59. Consider following reaction

$$\text{Chlorocyclohexane}(A) \xrightarrow{Cl_2/hv} \text{Dichlorocyclohexane}\ (B)$$

How many different isomers of B(geometrical and structural) will be produced?

60. (C – Cl) bond length in CH_3Cl is 178 pm. Its dipole moment is 1.87 D. Thus, percentage ionic character is……..%

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Mathematics

PART – C

SECTION – A

(Single Choice Answer Type)

This section contains **20 multiple choice questions**. Each question has four choices (A), (B), (C) and (D) out of which **ONLY ONE is correct.**

61. Suppose $\alpha \neq \frac{k\pi}{2}$ for any integer k and T = $\frac{\sin\alpha + \tan\alpha}{\cos\alpha + \cot\alpha}$. Which of the following statements is true?
(A) T takes negative values only
(B) T takes non-negative values only
(C) T takes positive values only
(D) T takes both positive and negative values

62. Suppose $\log_2[\log_{1/2}(\log_2 x)] = \log_3[\log_{1/3}(\log_3 y)] = \log_5[\log_{1/5}(\log_5 z)] = 0$. Which of the following statements is true ?
(A) $z < x < y$
(B) $x < y < z$
(C) $y < z < x$
(D) $z < y < x$

63. Suppose ϕ is any angle in the interval $\left(0, \frac{\pi}{2}\right)$. Which of the following statements is true ?
(A) $\sin\sin\phi < \cos\phi < \cos\cos\phi$
(B) $\sin\sin\phi > \cos\phi > \cos\cos\phi$
(C) $\sin\cos\phi > \cos\phi > \cos\sin\phi$
(D) $\sin\cos\phi < \cos\phi < \cos\sin\phi$

64. The function $f(x) = ax^2 - c$ satisfies $-4 \le f(1) \le -1$ and $-1 \le f(2) \le 5$. Which of the following statements is true ?
(A) $7 \le f(3) \le 26$
(B) $-4 \le f(3) \le 15$
(C) $-1 \le f(3) \le 20$
(D) $-\frac{28}{3} \le f(3) \le \frac{35}{3}$

65. Let $G(x) = \left(\frac{1}{a^x - 1} + \frac{1}{2}\right)F(x)$, where a is a positive real number not equal to 1 and F(x) is an odd function. Which of the following statements is true ?
(A) G(x) is an odd function
(B) G(x) is an even function
(C) G(x) is neither an odd function nor an even function
(D) Whether G(x) is an odd or even function depends on the value of a

66. In a tetrahedron, all sides are of length 1 except possibly one whose length is denoted by x. It volume is denoted by V(x). Which of the following statements is true ?
(A) V(x) is an increasing function which has no maximum
(B) V(x) is an increasing function which has a maximum
(C) V(x) is not an increasing function and has no maximum
(D) V(x) is not an increasing function but has a maximum

67. If none of the faces of a tetrahedron is an isosceles triangle, what is the minimum number of edges no two of which have the same length ?
(A) 3
(B) 4
(C) 5
(D) 6

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68. On the plane is point set M. For $1 \le i \le 7$, a circle C_i passes through exactly i points in M. What is the minimum number of points in M ?
(A) 11 (B) 12
(C) 21 (D) 28

69. Let α be a constant. What curve in the complex plane represents the set $\left\{-\bar{z}^2 \mid \arg z = \alpha\right\}$
(A) the ray arg z = 2α (B) the ray arg z = -2α
(C) the ray arg z = $-\alpha$ (D) none of (A), (B) and (C) is correct

70. For how many positive integer a < 100 will $a^3 + 23$ be divisible by 24 ?
(A) 4 (B) 5
(C) 9 (D) 10

71. Consider the function f(x) = $\frac{3x-1}{x^2+x}$. Which of the following is not correct ?
(A) f(x) is one-one $\forall\ x \in (1, \infty)$ (B) range of y = f(x) is (0, 1) if $x \in (1, \infty)$
(C) range of y = f(x) is [9, ∞) if $x \in (-1, 0)$ (D) f(x) is one-one $\forall\ x \in (-1, 0)$

72. Consider $f(x) = e^x + \int_0^x t.e^{t/x}dt$, then which of the following options are correct?
(A) range of y = f(x) is $(-\infty, \infty)$
(B) y = f(x) is one-one function
(C) there exist some $x \in (0, 1)$ such that f '(x) = e
(D) $\lim\limits_{x\to\infty} \frac{f(x)}{e^x} = 0$

73. Let p and q be two statements, then $(\sim p \vee q) \wedge (\sim p \wedge \sim q)$ is a
(A) tautology (B) contradiction
(C) neither tautology nor contradiction (D) can't say

74. The number of skew symmetric matrices of order 4×4, where entries belong to the set {–3, –2, –1, 0, 1, 2, 3} is
(A) 5^9 (B) 7^9
(C) 5^6 (D) 7^6

75. Solution of the differential equation $\left\{\frac{1}{x} - \frac{y^2}{(x-y)^2}\right\}dx + \left\{\frac{x^2}{(x-y)^2} - \frac{1}{y}\right\}dy = 0$ is
(A) $\log\left|\frac{x}{y}\right| + \frac{xy}{x-y} = c$ (B) $\frac{xy}{x-y} = c \cdot e^{x/y}$
(C) $\log|xy| = c + \frac{xy}{x-y}$ (D) $\log|xy| = c + \frac{1}{x-y}$

76. Let a and b be two positive real numbers, then the value of $I = \int_a^b \frac{\left(e^{x/a} - e^{b/x}\right)}{x}dx$ is
(A) ab (B) $\frac{1}{ab}$
(C) 1 (D) 0

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77. Let $\lim\limits_{x\to\pi}\frac{\sin\alpha x}{1-2\cos\beta x}=-\sqrt{3}$, $\alpha\in I^+$, $\beta>0$, then the least positive value of $\alpha\beta$ is

(A) $\frac{1}{6}$ (B) $\frac{1}{3}$

(C) $\frac{1}{2}$ (D) 1

78. Consider $f(x)=x^3+ax^2+bx+c$, parameters a, b and c are chosen respectively by throwing a dice three times. Then the probability that f(x) is an increasing function is

(A) $\frac{5}{36}$ (B) $\frac{8}{36}$

(C) $\frac{1}{3}$ (D) $\frac{4}{9}$

79. Let α and β are the eccentric angles of the end points of the focal chord of the hyperbola $\frac{x^2}{9}-\frac{y^2}{7}=1$, then the possible value of $\tan\frac{\alpha}{2}\cdot\tan\frac{\beta}{2}$ is

(A) -7 (B) 7

(C) $\frac{1}{5}$ (D) -5

80. If the expansion of $(1+x+x^2)^n$ is equal to $a_0+a_1x+a_2x^2+\ldots\ldots+a_{2n}x^{2n}$, the value of $\frac{a_0+a_1+a_3+a_4+a_6+a_7+a_9+a_{10}+\ldots\ldots}{a_2+a_5+a_8+a_{11}+\ldots\ldots}$ is

(A) $\frac{1}{3}$ (B) 2

(C) $\frac{1}{2}$ (D) 3

SECTION – B

(Numerical Answer Type)

This section contains **10** Numerical based questions. The answer to each question is rounded off to the nearest integer value.

81. Let $L_1: 2x+y-3=0$ and $L_2: x-2y+1=0$ be tangents to a parabola with focus S. If S_1 and S_2 be the image points of S about L_1 and L_2 respectively and mid-point of S_1 and S_2 is (α, β), then the value of $4\alpha-3\beta$ is equal to

82. Let the roots of the equation $x^4-4x^3+6x^2-4x-15=0$ are $\alpha, \beta, \gamma, \delta$, where $\alpha, \beta\in R$ then the value of $\gamma\delta-\alpha\beta$ is

83. Let shortest distance between a variable line and $\frac{x-1}{2}=\frac{y-2}{3}=\frac{z-3}{4}$ lies along the points A(3, 5, 7) and B(2, 3, 4), then the locus of the point in the variable line is $x+\alpha y+\beta z=20$ where $\alpha, \beta\in R$, then $\alpha+\beta$ is equal to

84. The sum of the series, $\cot^{-1}\left(1+\frac{3}{4}\right)+\cot^{-1}\left(2^2+\frac{3}{4}\right)+\cot^{-1}\left(3^2+\frac{3}{4}\right)+\ldots$ upto $\infty=\tan^{-1}(k)$, then the value of k is

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85. The number of points of discontinuity of f(x) = [5 – 4sinx], $x \in [0, \pi]$ where [.] is G.I.F. is

86. The area defined by $1 \le |x - 2| + |y + 1| \le 2$ is

87. $I_n = \int_0^{\infty} \frac{dx}{\left(x + \sqrt{x^2 + 1}\right)^n}$ where n > 1, then $[100I_8]$ (where [.] is G.I.F.) is equal to

88. If $\frac{1}{2}, \frac{1}{\alpha_1}, \frac{1}{\alpha_2}, \ldots\ldots\ldots \frac{1}{\alpha_{20}}, \frac{1}{6}$ are in A.P. and 2, β_1, β_2, …… , β_{20}, 6 are in A.P., then the value of $\frac{\alpha_{18} \cdot \beta_3}{2}$ is

89. If $\vec{a}$ and $\vec{b}$ are any two unit vectors, then the maximum possible integral value of $3\left|\vec{a} - \vec{b}\right| + \frac{7}{2}\left|\vec{a} + \vec{b}\right|$ is

90. $\int e^x \left(x^5 - 4x^4 + 3x^2 + 2\right)dx = \left(x^5 + \alpha x^4 + \beta x^3 + \gamma x^2 + \gamma x + p\right)e^x + c$, then $\alpha + \beta$ is

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